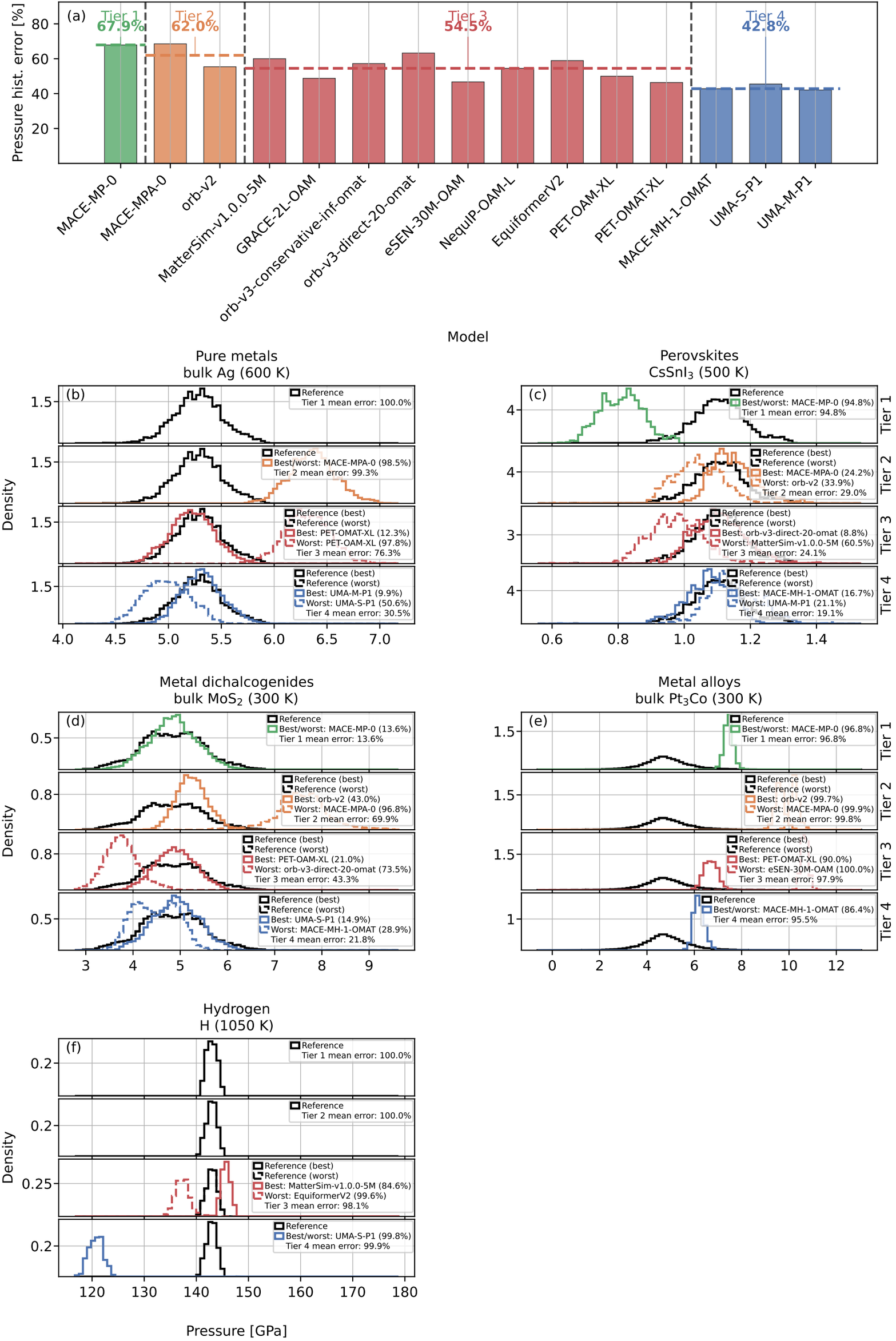

(a)
Tier 1
67.9%
Tier 2
62.0%
Tier 3
54.5%
Tier 4
42.8%
Pressure hist. error [%]
MACE-MP-0
MACE-MPA-0
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-OAM
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1
UMA-M-P1
Model
Pure metals
bulk Ag (600 K)
(b)
Reference
Tier 1 mean error: 100.0%
Best/worst: MACE-MPA-0 (98.5%)
Tier 2 mean error: 99.3%
Reference (best)
Reference (worst)
Best: PET-OMAT-XL (12.3%)
Worst: PET-OAM-XL (97.8%)
Tier 3 mean error: 76.3%
Best: UMA-M-P1 (9.9%)
Worst: UMA-S-P1 (50.6%)
Tier 4 mean error: 30.5%
Density
Perovskites
CsSnI3 (500 K)
(c)
Best/worst: MACE-MP-0 (94.8%)
Tier 1 mean error: 94.8%
Best: MACE-MPA-0 (24.2%)
Worst: orb-v2 (33.9%)
Tier 2 mean error: 29.0%
Best: orb-v3-direct-20-omat (8.8%)
Worst: MatterSim-v1.0.0-5M (60.5%)
Tier 3 mean error: 24.1%
Best: MACE-MH-1-OMAT (16.7%)
Worst: UMA-M-P1 (21.1%)
Tier 4 mean error: 19.1%
Metal dichalcogenides
bulk MoS2 (300 K)
(d)
Best/worst: MACE-MP-0 (13.6%)
Tier 1 mean error: 13.6%
Best: orb-v2 (43.0%)
Worst: MACE-MPA-0 (96.8%)
Tier 2 mean error: 69.9%
Best: PET-OAM-XL (21.0%)
Worst: orb-v3-direct-20-omat (73.5%)
Tier 3 mean error: 43.3%
Best: UMA-S-P1 (14.9%)
Worst: MACE-MH-1-OMAT (28.9%)
Tier 4 mean error: 21.8%
Metal alloys
bulk Pt3Co (300 K)
(e)
Best/worst: MACE-MP-0 (96.8%)
Tier 1 mean error: 96.8%
Best: orb-v2 (99.7%)
Worst: MACE-MPA-0 (99.9%)
Tier 2 mean error: 99.8%
Best: PET-OMAT-XL (90.0%)
Worst: eSEN-30M-OAM (100.0%)
Tier 3 mean error: 97.9%
Best/worst: MACE-MH-1-OMAT (86.4%)
Tier 4 mean error: 95.5%
Hydrogen
H (1050 K)
(f)
Tier 1 mean error: 100.0%
Tier 2 mean error: 100.0%
Best: MatterSim-v1.0.0-5M (84.6%)
Worst: EquiformerV2 (99.6%)
Tier 3 mean error: 98.1%
Best/worst: UMA-S-P1 (99.8%)
Tier 4 mean error: 99.9%
Pressure [GPa]